

Assignment 3: Data Exploration

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Exploration.

Directions

1. Rename this file `<FirstLast>_A03_DataExploration.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Assign a useful **name to each code chunk** and include ample **comments** with your code.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.
7. After Knitting, submit the completed exercise (PDF file) to the dropbox in Canvas.

TIP: If your code extends past the page when knit, tidy your code by manually inserting line breaks.

TIP: If your code fails to knit, check that no `install.packages()` or `View()` commands exist in your code.

Set up your R session

1. Load necessary packages (tidyverse, lubridate, here), check your current working directory and upload two datasets: the ECOTOX neonicotinoid dataset (ECOTOX_Neonicotinoids_Insects_raw.csv) and the Niwot Ridge NEON dataset for litter and woody debris (NEON_NIWO_Litter_massdata_2018-08_raw.csv). Name these datasets “Neonics” and “Litter”, respectively. Be sure to include the sub-command to read strings in as factors.

```
# Setting up packages and working directory

# Loading packages
library(tidyverse); library(lubridate); library(here)

# Checking working directory
getwd()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
here()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
# Uploading two data sets
```

```
Neonics <- read.csv(  
  here("./Data/Raw/ECOTOX_Neonicotinoids_Insects_raw.csv"),  
  stringsAsFactors = TRUE  
)  
  
Litter <- read.csv(  
  here("./Data/Raw/NEON_NIW0_Litter_massdata_2018-08_raw.csv"),  
  stringsAsFactors = TRUE  
)
```

Learn about your system

2. The neonicotinoid dataset was collected from the Environmental Protection Agency's ECOTOX Knowledgebase, a database for ecotoxicology research. Neonicotinoids are a class of insecticides used widely in agriculture. The dataset that has been pulled includes all studies published on insects. Why might we be interested in the ecotoxicology of neonicotinoids on insects? Feel free to do a brief internet search if you feel you need more background information.

Answer: Insects play a large role in agricultural systems. They can be both harmful and beneficial from the standpoint of agricultural output and overall ecosystem health. Therefore, it would be important to understand the potential impacts that neonicotinoids have on insects. We'd be interested in information such as which insects the insecticides are impacting. Are the insecticides acting as intended? Are there other unintended, potentially harmful consequences from their use? Answering these kinds of questions would be very important for a range of stakeholders in the agricultural/insecticide space/industries.

3. The Niwot Ridge litter and woody debris dataset was collected from the National Ecological Observatory Network, which collectively includes 81 aquatic and terrestrial sites across 20 ecoclimatic domains. 32 of these sites sample forest litter and woody debris, and we will focus on the Niwot Ridge long-term ecological research (LTER) station in Colorado. Why might we be interested in studying litter and woody debris that falls to the ground in forests? Feel free to do a brief internet search if you feel you need more background information.

Answer: Forest litter and woody debris are key components in long-term forest and ecological processes. The non-living matter eventually decomposes and their nutrients are returned to the soil, which then supports other plant/animal life. Studying patterns in the amounts and distributions of forest litter and woody debris might reveal interesting insights. For example, is climate change impacting certain areas and the forests' cycle more than others? Are there increases/decreases in the amount of litter? Understanding/revealing any potential patterns matter given the importance of these components in ecosystem health and what they can reveal about how well a forest is functioning.

4. How is litter and woody debris sampled as part of the NEON network? Read the `NEON_Litterfall_UserGuide.pdf` document to learn more. List three pieces of salient information about the sampling methods here:

Answer: 1. Litter and fine woody debris are collected from elevated and ground traps. 2. Sampling takes place at terrestrial NEON (National Ecological Observatory Network) sites that contain woody vegetation over 2 meters tall. 3. Ground traps are sampled once per year. Elevated traps are sampled at various frequencies depending on the vegetation at the site (once every 2 weeks in deciduous sites and once every 1-2 months at evergreen sites).

Obtain basic summaries of your data (Neonics)

5. What are the dimensions of the dataset?

```
# Obtaining the dimensions of the Neonics dataset
```

```
dim(Neonics)
```

```
## [1] 4623 30
```

6. Using the `summary` function on the “Effect” column, determine the most common effects that are studied. Why might these effects specifically be of interest? [Tip: The `sort()` command is useful for listing the values in order of magnitude...]

```
# Identifying the effects and listing them by order of magnitude
```

```
sort(summary(Neonics$Effect))
```

```
##      Hormone(s)      Histology      Physiology      Cell(s)
##           1           5           7           9
##      Biochemistry      Accumulation      Intoxication      Immunological
##          11          12          12          16
##      Morphology      Growth      Enzyme(s)      Genetics
##          22          38          62          82
##      Avoidance      Development      Reproduction      Feeding behavior
##         102         136          197          255
##      Behavior      Mortality      Population
##         360         1493          1803
```

Answer: The two most common effects are population and mortality. These are of interest because in agricultural systems, the population/presence of certain insect species matters greatly. For example, farmers would want to know if the population of pests is being impacted by insecticides and in what way. And they’d also want to know if the population of pollinators is being impacted. Are populations growing or shrinking? Are certain species dying off? These are all important questions in contexts such as this one.

7. Using the `summary` function, determine the six most commonly studied species in the dataset (common name). What do these species have in common, and why might they be of interest over other insects? Feel free to do a brief internet search for more information if needed.[TIP: Explore the help on the `summary()` function, in particular the `maxsum` argument...]

```
# Identifying the six most commonly studied species for Species.Common.Name
```

```
summary(Neonics$Species.Common.Name, maxsum = 6)
```

##	Honey Bee	Parasitic Wasp	Buff Tailed Bumblebee
##	667	285	183
##	Carniolan Honey Bee	Bumble Bee	(Other)
##	152	140	3196

Answer: The most common species in the dataset are all pollinators. The ‘Other’ category is also quite large. These would be of interest over other insects given the important role they play in enabling plants to produce fruits and other forms of food.

8. Concentrations are always a numeric value. What is the class of `Conc.1..Author.` column in the dataset, and why is it not numeric? [Tip: Viewing the dataframe may be helpful...]

```
#Identifying the class of Conc.1..Author
```

```
class(Neonics$Conc.1..Author.)
```

```
## [1] "factor"
```

Answer: It’s not numeric as there are other non-numeric values present. Some cells are notated with a / or >. NR is also another value. To account for all these, it makes sense to have the class be factor rather than numeric.

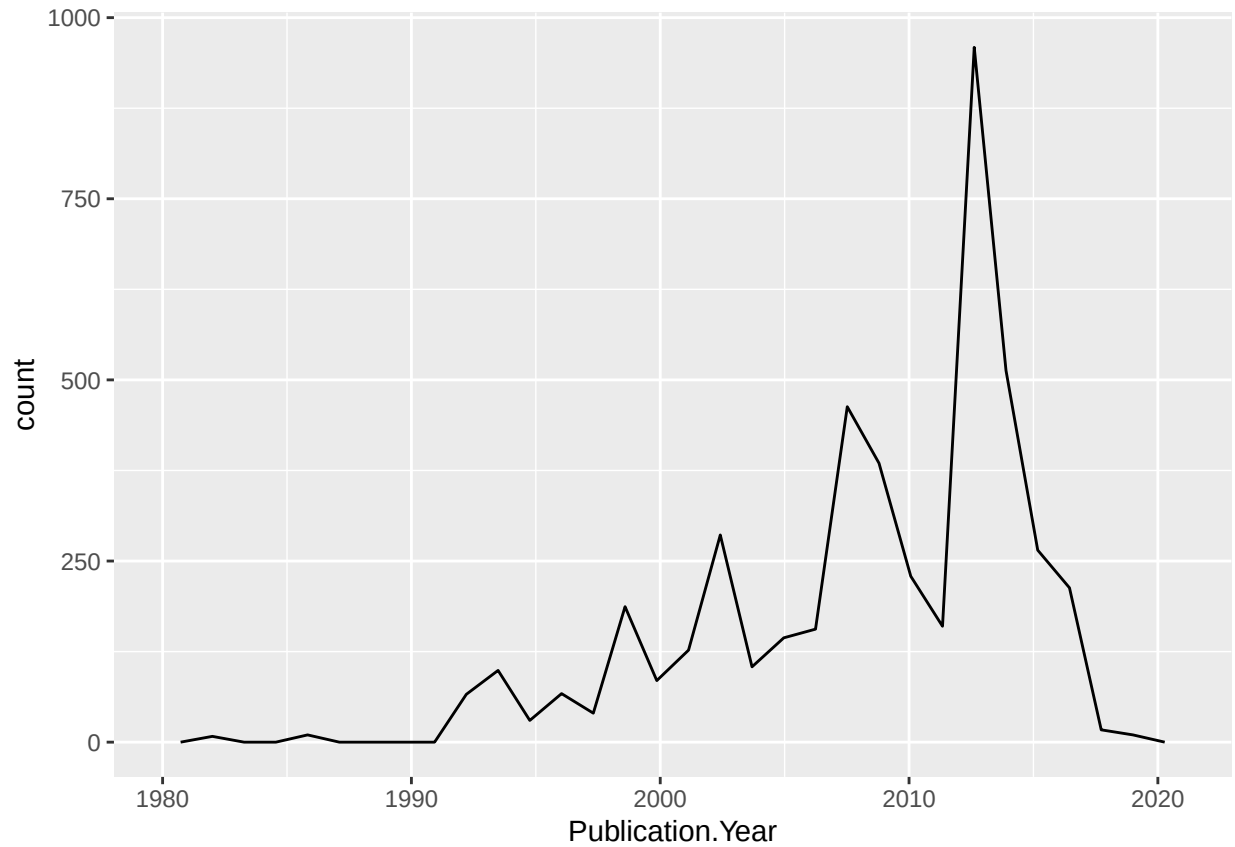
Explore your data graphically (Neonics)

9. Using `geom_freqpoly`, generate a plot of the number of studies conducted by publication year.

```
# Making a line plot of the number of studies by publication year
```

```
ggplot(data=Neonics) +  
  geom_freqpoly(aes(x=Publication.Year))
```

```
## ‘stat_bin()’ using ‘bins = 30’. Pick better value ‘binwidth’.
```

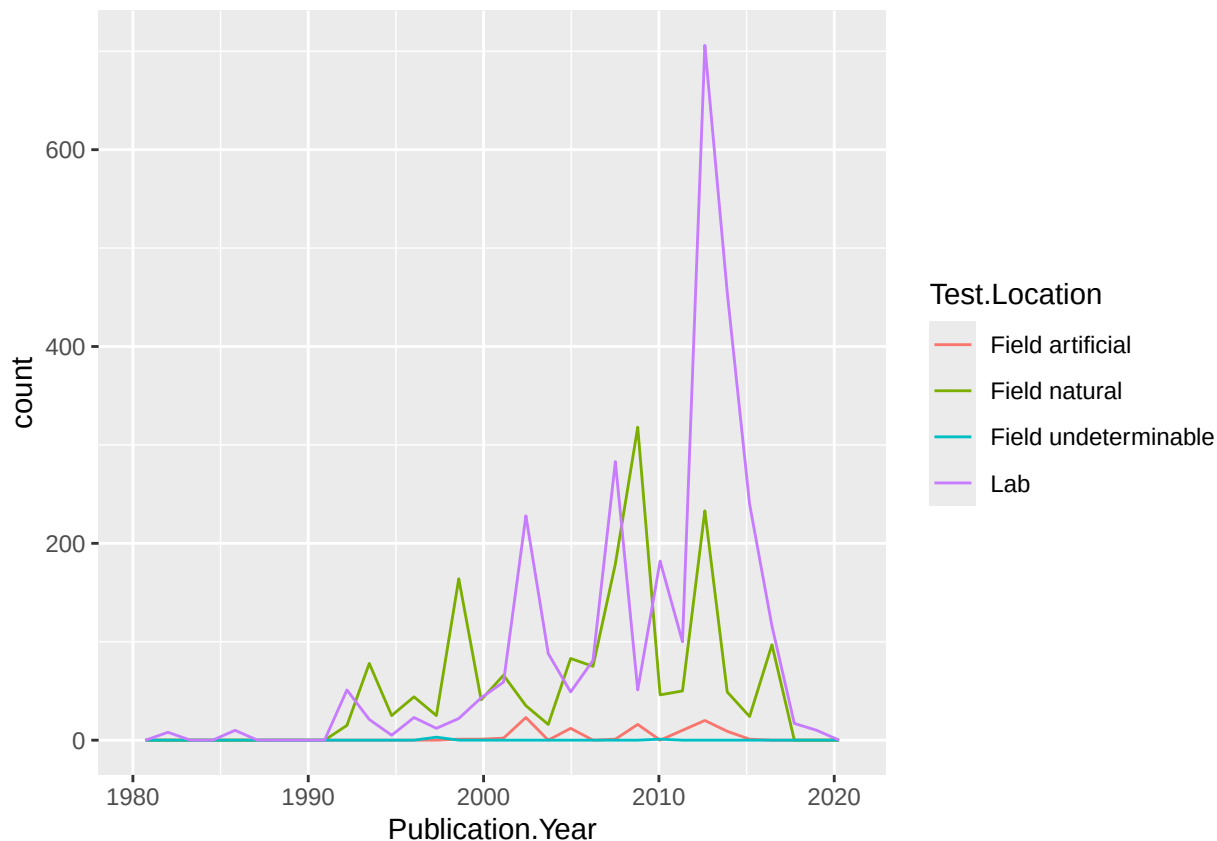


10. Reproduce the same graph but now add a color aesthetic so that different Test.Location are displayed as different colors.

```
# Creating the same plot and adding in test location by color

ggplot(data=Neonics) +
  geom_freqpoly(aes(x=Publication.Year, color = Test.Location))

## 'stat_bin()' using 'bins = 30'. Pick better value 'binwidth'.
```



Interpret this graph. What are the most common test locations, and do they differ over time?

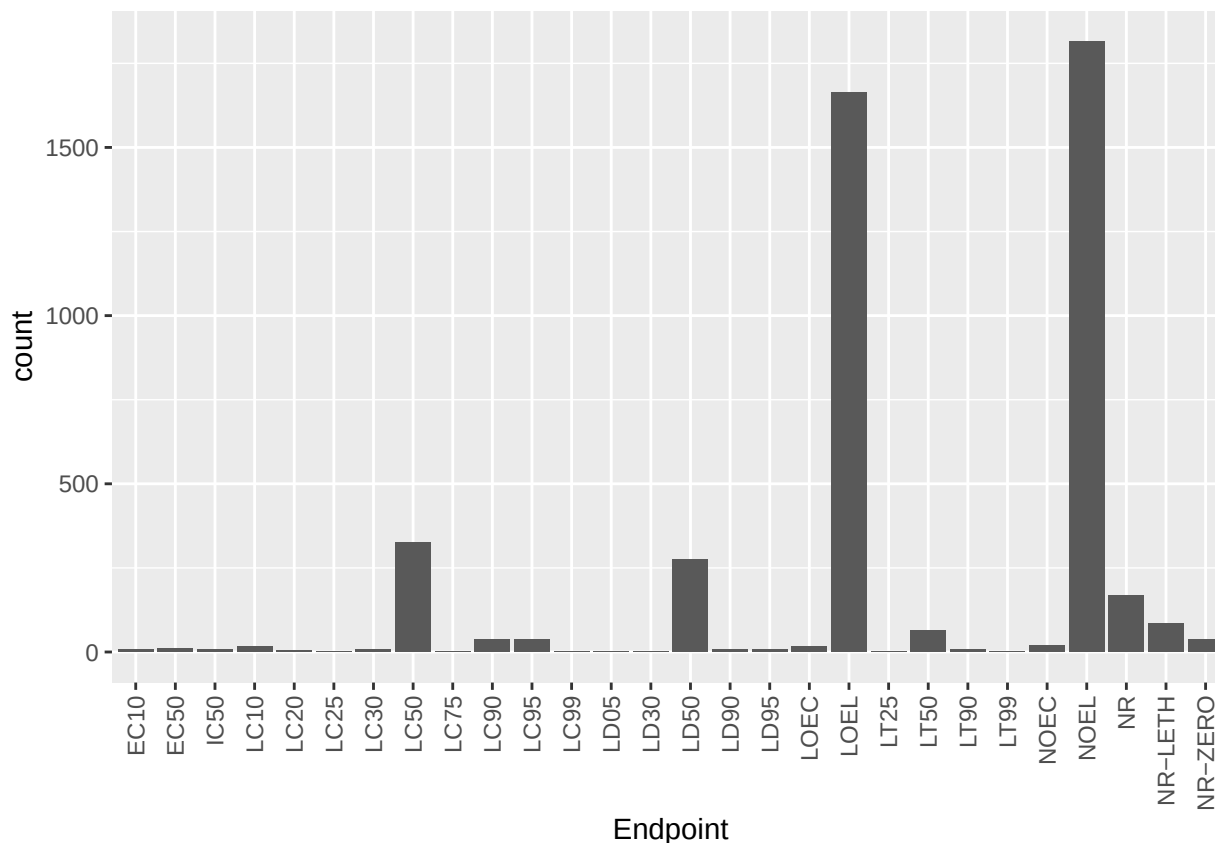
Answer: The two most common test locations are 'Lab' and 'Field natural.' It seems from about 1990 to 2000, more tests were conducted in field natural locations, and then starting around 2000, lab became the more dominant location with a very big surge between 2010 and 2015.

11. Create a bar graph of Endpoint counts. What are the two most common end points, and how are they defined? Consult the ECOTOX_CodeAppendix for more information.

[TIP: Add `theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))` to the end of your plot command to rotate and align the X-axis labels...]

Creating a bar graph of Endpoint counts

```
ggplot(data=Neonics, aes(x=Endpoint)) +
  geom_bar() +
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))
```



Answer: The two most common endpoints are NOEL and LOEL. They both have terrestrial database usages. NOEL is defined as “no-observable-effect-level,” with the highest dose producing statistically insignificant effects. LOEL is defined as “lowest-observable-effect-level,” with the lowest concentration producing statistically significant effects.

Explore your data (Litter)

- Determine the class of collectDate. Is it a date? If not, change to a date and confirm the new class of the variable. Using the `unique` function, determine which dates litter was sampled in August 2018.

```
class(Litter$collectDate) # Checking the class of collectDate
```

```
## [1] "factor"
```

```
Litter$collectDate <- ymd(Litter$collectDate) #Changing the class of collectDate
class(Litter$collectDate) # Confirming change was successful
```

```
## [1] "Date"
```

```
unique(Litter$collectDate) # Seeing which dates litter was sampled in August 2018
```

```
## [1] "2018-08-02" "2018-08-30"
```

13. Using the `unique` function, list the different plotIDs sampled at Niwot Ridge. How is the information obtained from `unique` different from that obtained from `summary`?

```
unique(Litter$plotID)
```

```
## [1] NIWO_061 NIWO_064 NIWO_067 NIWO_040 NIWO_041 NIWO_063 NIWO_047 NIWO_051
## [9] NIWO_058 NIWO_046 NIWO_062 NIWO_057
## 12 Levels: NIWO_040 NIWO_041 NIWO_046 NIWO_047 NIWO_051 NIWO_057 ... NIWO_067
```

```
summary(Litter$plotID)
```

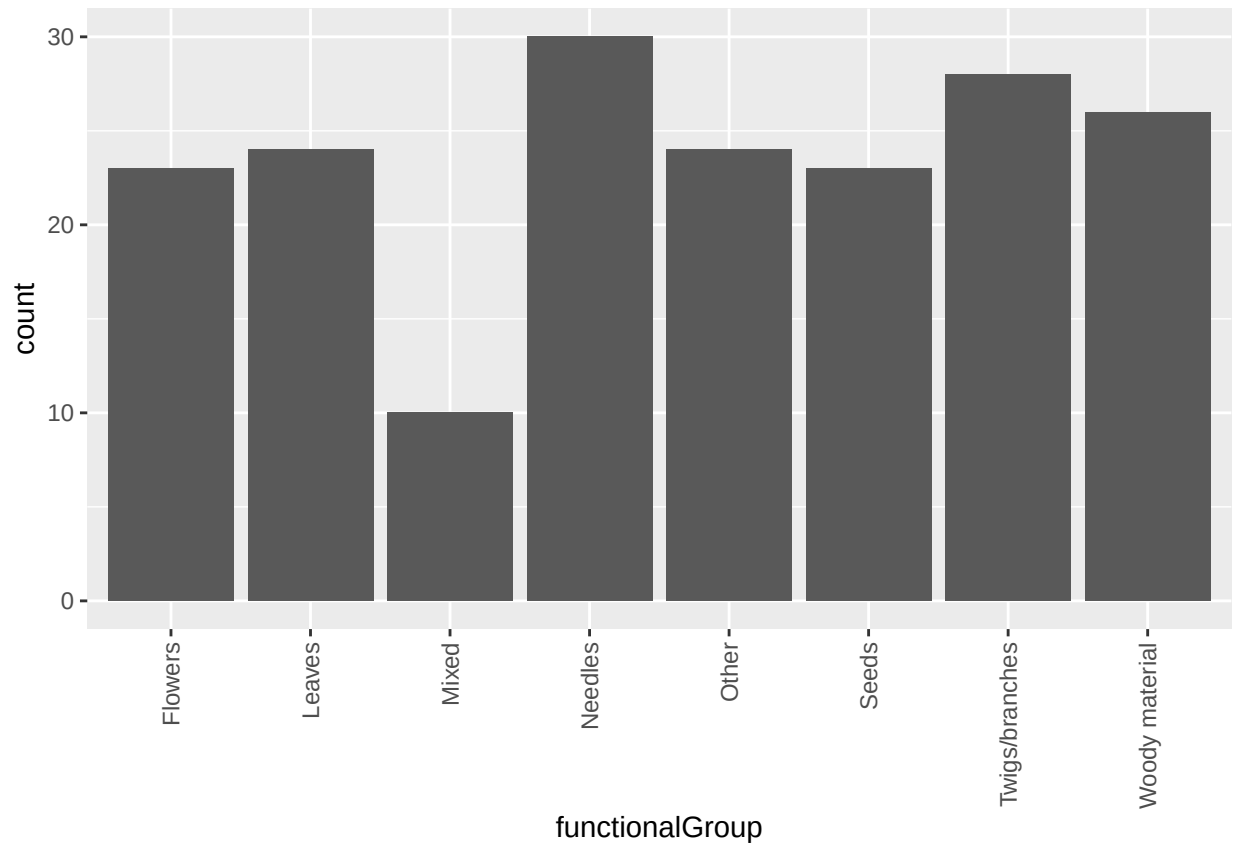
```
## NIWO_040 NIWO_041 NIWO_046 NIWO_047 NIWO_051 NIWO_057 NIWO_058 NIWO_061
##      20      19      18      15      14      8      16      17
## NIWO_062 NIWO_063 NIWO_064 NIWO_067
##      14      14      16      17
```

Answer: In addition to listing the different plotIDs like the `unique` command does, the `summary` command also provides how many times each appears in the dataset. The `unique` command does provide the number of levels too though, which the `summary` command does not.

14. Create a bar graph of functionalGroup counts. This shows you what type of litter is collected at the Niwot Ridge sites. Notice that litter types are fairly equally distributed across the Niwot Ridge sites.

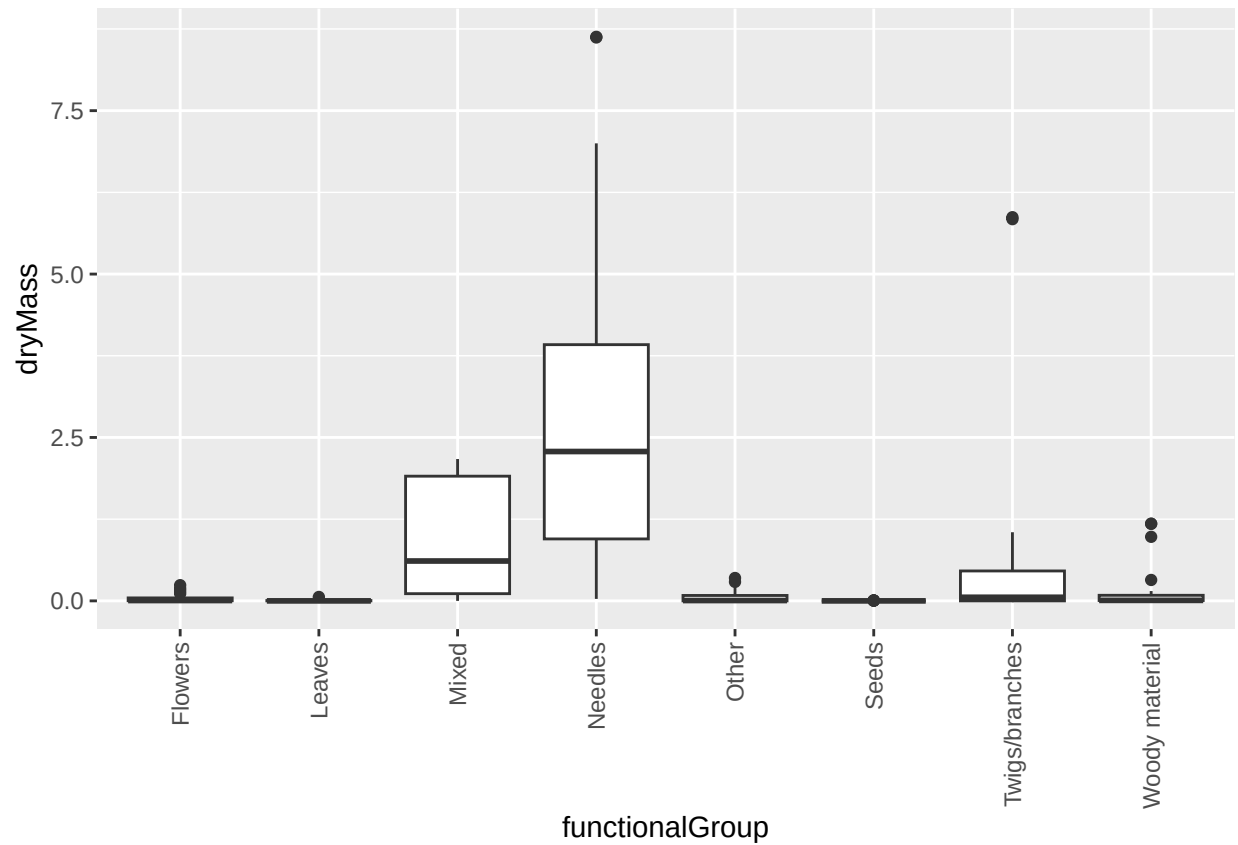
```
# Creating a bar graph of functionalGroup
```

```
ggplot(data = Litter, aes(x = functionalGroup)) +
  geom_bar() +
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))
```

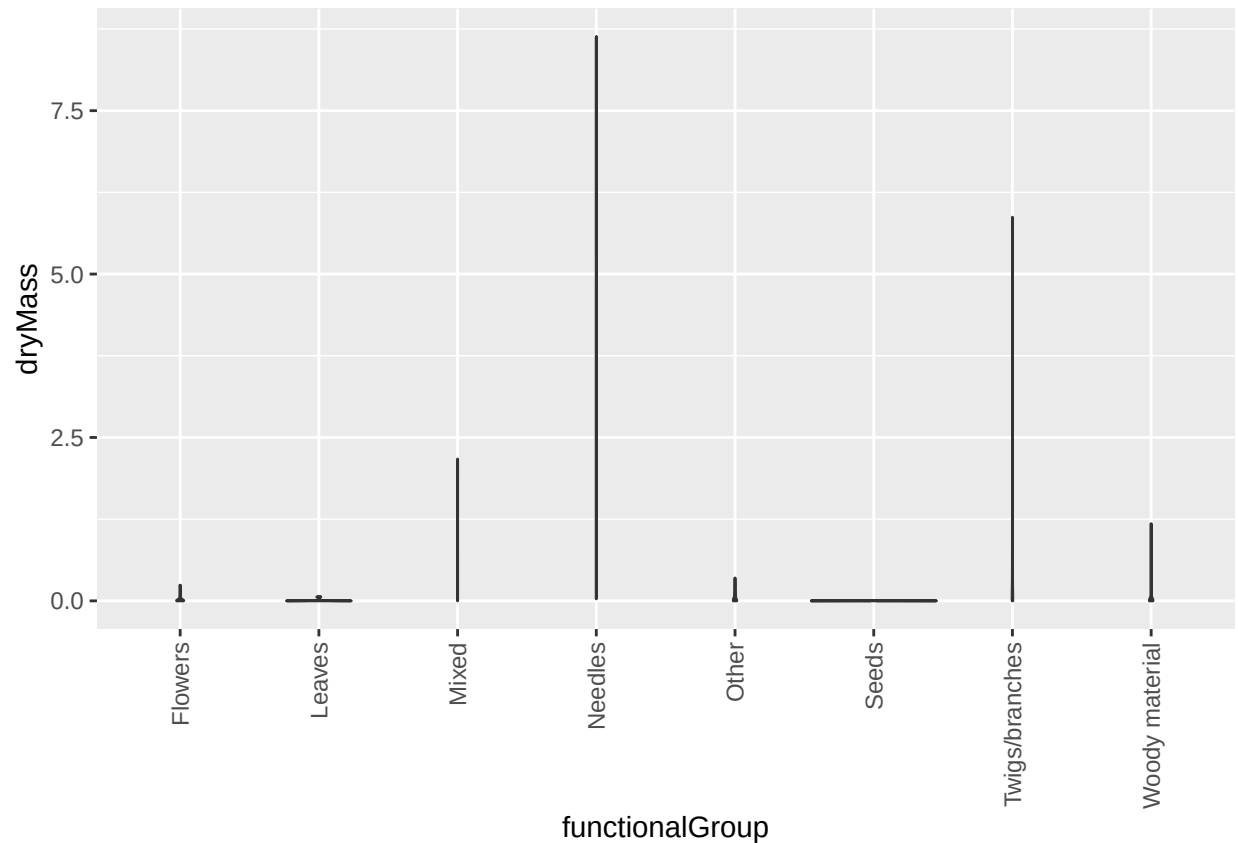
15. Using `geom_boxplot` and `geom_violin`, create a boxplot and a violin plot of `dryMass` by `functionalGroup`.

```
# Creating a boxplot  
ggplot(data = Litter) +  
  geom_boxplot(aes(x = functionalGroup, y = dryMass)) +  
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))
```



Creating a violin plot

```
ggplot(data = Litter) +  
  geom_violin(aes(x = functionalGroup, y = dryMass)) +  
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))
```



Why is the boxplot a more effective visualization option than the violin plot in this case?

Answer: The violin plot results in just straight lines. I believe this means that there is little to no variance in the values. Additionally, this may also be happening as the dataset is fairly small, so the added density layer isn't coming through very well. As such, the boxplot is a more effective visualization given its simplicity.

What type(s) of litter tend to have the highest biomass at these sites?

Answer: Needles and the 'mixed' category have the highest biomass at these sites.